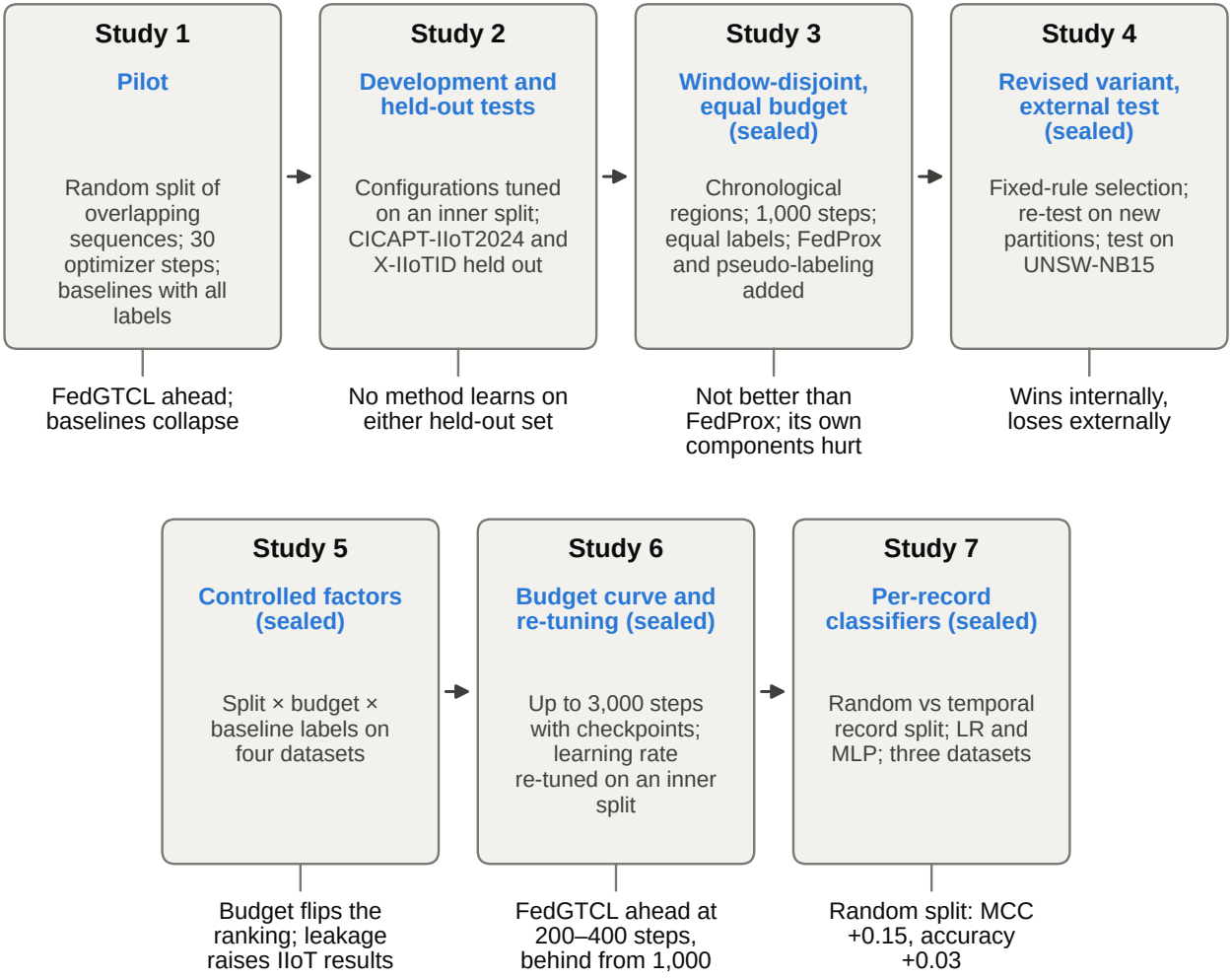
Study 1
Pilot
Random split of overlapping sequences; 30 optimizer steps; baselines with all labels
FedGTCL ahead; baselines collapse
Study 2
Development and held-out tests
Configurations tuned on an inner split; CICAPT-IIoT2024 and X-IIoTID held out
No method learns on either held-out set
Study 3
Window-disjoint, equal budget (sealed)
Chronological regions; 1,000 steps; equal labels; FedProx and pseudo-labeling added
Not better than FedProx; its own components hurt
Study 4
Revised variant, external test (sealed)
Fixed-rule selection; re-test on new partitions; test on UNSW-NB15
Wins internally, loses externally
Study 5
Controlled factors (sealed)
Split × budget × baseline labels on four datasets
Budget flips the ranking; leakage raises IIoT results
Study 6
Budget curve and re-tuning (sealed)
Up to 3,000 steps with checkpoints; learning rate re-tuned on an inner split
FedGTCL ahead at 200–400 steps, behind from 1,000
Study 7
Per-record classifiers (sealed)
Random vs temporal record split; LR and MLP; three datasets
Random split: MCC +0.15, accuracy +0.03